

The Aperiodic Holography Theorem for Spectre Monotiles

Formally Verifying 2D Bulk Rigidity from 1D Boundary Strings in Lean 4

James F. Walters

Port Townsend Recreational Math Club

May 25, 2026

The Hook: Reducing Tiling Dimensionality

- **The Discovery:** The Spectre monotile establishes a strictly chiral Einstein parameter space—enforcing non-periodic planar tilings under translation and rotation without reflections.
- **The Geometric Obstacle:** Traditional multi-tile patch validation requires continuous 2D planar graph tracking, ray-casting intersection checks, and float-heavy coordinate sweeps.
- **Our Formulation:** The **Aperiodic Holography Theorem** completely bypasses multi-dimensional search fields.
- **The Core Thesis:** The 1D sequence of exterior boundary turns enclosing a finite, simply connected patch uniquely and deterministically dictates the full 2D interior configuration.

The Holographic Paradigm

The boundary path string entirely determines the bulk. Distinct perimeters map injectively to distinct interiors.

Formalizing the 1D Lattice Walk

Each boundary step along the path is serialized as an inductive structure tracking local and absolute geometric invariants:

The BoundaryStep Structure (Lean 4)

- `dir : EdgeDirection` $\implies$ Absolute compass heading mapped flatly to $\mathbb{Z}_{12}$ (Fin 12) via 30° discretization steps.
- `turn : ExteriorTurn` $\implies$ Localized change in direction matching one of five discrete rotational lattice vertices.
- `parity : EdgeParity` $\implies$ Alternating binary tracking state mapping the current edge segment to its structural type.

$$\text{Permissible Turns} \in \{t_{-90}, t_{-60}, t_0, t_{60}, t_{90}\}$$

The Curvature Constraint (Discrete Gauss-Bonnet)

A valid path traversing a closed counterclockwise boundary loop must accumulate a total net rotation of exactly $+360^\circ$ to achieve valid loop closure.

- **The Diophantine Turning Equation:** Dividing the loop rotation by our scale factor of 30° yields an absolute algebraic invariant:

$$3L_{90} + 2L_{60} - 2R_{60} - 3R_{90} = 12 \quad (1)$$

- **The Edge Parity Constraint:** Because 60° steps preserve parity and 90° steps toggle it, returning to the starting segment requires an even number of orthogonal transitions:

$$(L_{90} + R_{90}) \equiv 0 \pmod{2} \quad (2)$$

The Convex Anchor Theorem

Theorem (Convex Anchor Necessity)

Every valid, non-empty boundary path B enclosing a multi-tile bulk patch mathematically requires at least one academic, outward-pointing left 90° corner ($L_{90} \geq 1$).

Proof Sketch by Contradiction:

- 1 Assume $L_{90} = 0 \implies R_{90}$ is an even integer $2k$ via parity constraint.
- 2 Substituting into the Turning Equation yields: $L_{60} = 6 + R_{60} + 3k$.
- 3 This forces a major surplus of Left 60° transitions ($L_{60} \geq 6$), forming a wrapped hull that traps internal vertices.
- 4 The massive pool of interior 90° corners ($5N$ corners scaling against an $O(\sqrt{N})$ boundary capacity) cannot escape, forcing them to assemble into internal four-tile crosses.

The Cyclotomic Integer Lattice Contradiction

To eliminate floating-point rounding errors, all geometry is performed over the ring of cyclotomic integers $\mathbb{Z}[\zeta_{12}]$ as a free $\mathbb{Z}$ -module of rank 4: $(a, b, c, d) \in \mathbb{Z}^4$.

To evaluate segment collisions exactly, points project into the quadratic integer sub-ring $\mathbb{Z}[\sqrt{3}] \subset \mathbb{R}$ via integer scaling:

$$\text{toPoint2D}(a, b, c, d) = \left((2a + c) + b\sqrt{3}, (b + 2d) + c\sqrt{3} \right)$$

```
-- Exact compile-time evaluation of all 625 cross orientations
theorem crosses_always_overlap :  $\forall c \in \text{generateCrosses}, \text{checkCrossOverlap } c =$ 
  true := by
  decide
```

Contradiction Closed

The Lean kernel certifies that all 625 cross configurations result in physical overlaps. Internal crosses cannot exist $\implies L_{90} \geq 1$.

Chiral Forcing and the 155-Rule FSM

- Because the Spectre monotile is strictly chiral and asymmetric, no two 90° corners on a single tile share the same flanking turns.
- Minimum interior angles are 90° , meaning exactly one tile can occupy a convex corner vertex without causing overlap violations.

```
-- Computational proof that all 3-turn corner profiles are unique
theorem spectre_corners_are_unique : (getTileTriplets spectrePerimeterTurns).
  Nodup := by
  decide
```

Reading the 3-turn sliding window centered on a convex anchor uniquely forces the identity and orientation of that tile, matching a static lookup database of **155 deterministic prefix rewrite rules**.

The Peeling Operator and Seam Welding

The reduction of a patch perimeter is governed by the core single-tile peeling operator `peelBoundary` inside `SpectreBoundary.lean`:

- 1 **Cropping:** Rotates the list to the anchor index and drops matching exposed external turns via `List.drop rule.pattern.length`.
- 2 **Splicing:** Generates and splices in the inverted sequence of unexposed interior steps via `propagateSplicedSteps`.
- 3 **Stitch Welding:** Calculates and updates interface headings at connection joints via `steps_updated` and `updateLastTurn` to preserve directional alignment.

Certified Boundary Conservation

The Lean kernel formally certifies that this list surgery preserves macroscopic consistency: `peelBoundary_dir_consistent` verifies heading alignment and loop curvature closure across connection seams.

Well-Founded Global Uniqueness

The recursive decoder dynamically targets a convex anchor, maps the forced neighborhood, and peels the boundary to establish a well-founded reduction loop:

```
noncomputable def decodeInterior (B : BoundaryPath) : List TileId :=
  if B.tile_count <= 1 then []
  else
    let i := Classical.choose (convex_anchor_index B)
    let T := Classical.choose (forcing_neighborhood B i ...)
    match h_peel : peelBoundary B i with
    | some B' => T :: decodeInterior B'
    | none    => [T]
termination_by B.tile_count
```

Because each peel strictly decreases the patch area ($B'.tile_count < B.tile_count$), Lean certifies termination and closes global uniqueness:

`theorem aperiodic_holography_uniqueness` $\implies$ The 1D boundary path string uniquely determines the 2D interior tile bulk.